The NEXUS Quantum Compiler

Mathematical-Space Execution and Constrained Molecular Folding

Native reversible routes · quantum-sector reduction · legal molecular transformations · classical execution

Driven by Dean A. Kulik ·
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ABSTRACT

A quantum-compatible mathematical state space can be compiled and executed on ordinary classical hardware when its constraints, return paths, and observable symmetry sectors are explicitly represented. This manuscript integrates two complementary NEXUS studies without replacing either with a shorter summary. The first examines the exact native ring and E/G return machinery and audits an earlier quantum lattice-protein simulation whose eight-address mixer introduced four transitions through invalid padding states. It derives the short-time leakage coefficient 49/75, explains the distinction between finite penalties and exact subspace preservation, identifies the disconnected projected graph, and records the approximation and environmental-return requirements. The second advances to a self-avoiding honeycomb-chain compiler in which turn words construct geometry and admissible bit rewrites construct the transition graph. Global complement is an exact mirror symmetry; for a complement-stable Hamiltonian it separates the quantum space into even and odd sectors, retaining the necessary sheet sign and inverse readout. In the 12-residue HP example, 776 legal conformations form 388 mirror pairs. Classical propagation in the full Hilbert space and its independently assembled mirror-even sector agrees after reconstruction to machine precision under the reported sparse solver; the precise rounding error is implementation dependent. Local four-turn hexagon rewrites are reversible but fragment the legal graph; unrestricted pivots reconnect it while sacrificing bounded spatial support. Together these results establish a rigorous, finite compiler architecture and a precise reduction in the resources needed for symmetry-selected states. They do not establish experimentally validated molecular kinetics, a general quantum-simulation speedup, or measured low electrical power. The remaining link is an independently constrained molecular transition generator, its environmental coupling, and a resource benchmark carried out under the same input–output contract.

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Keywords: NEXUS; reversible routing; quantum-inspired classical simulation; parity decomposition; Hilbert-space reduction; self-avoiding walks; hydrophobic-polar models; constrained transitions; local moves; protein folding.

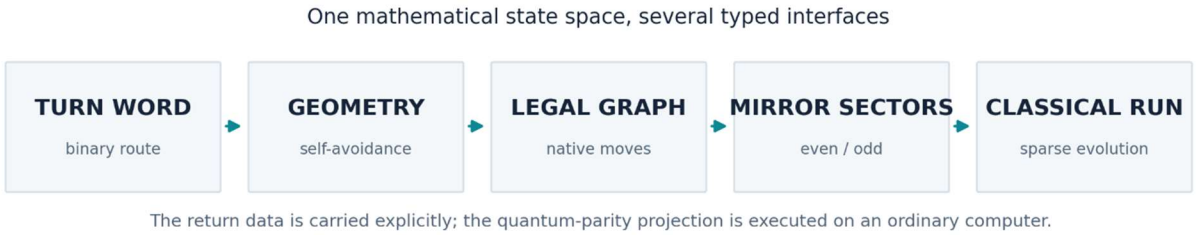


Figure 1. The integrated compiler path: binary word → geometric legality → admissible transition graph → symmetry sectors → classical propagation. The route back to full state is explicit.

PART I / THE NATIVE STATE SPACE AND WHAT "QUANTUM COMPILER" MEANS

1. Problem statement: computation before hardware

The motivating hypothesis is that a sufficiently explicit mathematical relation space can execute useful quantum-style calculations on conventional, low-power hardware. In its strongest form, this would make special-purpose quantum machines unnecessary for the targeted tasks. A physically meaningful version of that hypothesis must distinguish three different propositions: an algebraic operator can be represented classically; a specified quantum evolution can be simulated faithfully; and the simulation has an advantageous cost in time, memory, or joules. The first two are demonstrated here in a finite structured family. The third remains subject to benchmarking.

The Nexus program approaches problems from legal transformations and their return paths rather than from cataloged answers. Earlier One Inverter and twisted-partner-ring work constructed involutions, native addresses, temporal center reads, and odd-lane locks [1, 2]. The honeycomb extensions (§§144–146) supply a particularly instructive domain: a word is a conformation-generating route, while a molecular move is determined by geometry rather than by neighboring values in an address table [2].

Classical simulation of quantum subfamilies is established mathematical territory. Stabilizer circuits admit efficient classical simulation [6], and quantum-inspired linear algebra can dequantize some low-rank algorithms under explicit data-access assumptions [7, 8]. The point of the present work is narrower: show where the Nexus native-route and reflection machinery gives a concrete geometry-respecting Hilbert-space reduction, with complete return information and negative controls.

The governing research hypothesis is stronger than an indexing engine. The compiler is proposed as a transformation architecture: a molecular sequence and surroundings determine lawful continuations, and a fold is the realized trajectory rather than a catalog match. The scientific question is whether a native, local, reversible transformation system reproduces a molecular process without placing the answer in a chosen transition matrix.

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Three different "one answer" statements must remain typed. A fully specified deterministic calculation has one result. A fully specified quantum experiment has a definite predicted probability distribution but may have multiple measurement outcomes. A protein may have an ensemble of conformations that changes with conditions. The route does not license substituting one for the others.

For this paper, "compilation" has a testable meaning: construct a state domain, certify lawful transformations, select a representation that makes required relations visible, propagate that declared operator, and preserve an inverse or a residual route. Classical hardware is sufficient to carry out these finite algebraic steps. Whether it is efficient for an enlarging family depends on the structure of that family, not on a claim that all quantum amplitudes can be recovered from one address.

2. Native ring: partner, face, canonical address, and exact return

For an n-bit word, J is bit reversal and C is bitwise complement. Define P(x)=J(x) unless x is palindromic, in which case P(x)=C(x). $P^2=I$, P has no fixed word, and the native ring pairs addresses antipodally: $PTP=T^{-1}$, where T is the ring's own successor. The inward E/G route ends at width two, retaining each branch and middle qualifier and a face bit. The complete code is n bits for an n-bit word; reconstruction is exact but not generic compression.

$P(x) = J(x)$ if $J(x) \neq x$; $P(x) = C(x)$ otherwise.

The canonical address is an exact re-labeling of word space. T is a native address successor. A raw k-bit transport shift, a partner read, and T are separate operators. In particular, declaring an edge between successive catalog addresses does not establish that the corresponding conformations are molecular neighbors.

Three interfaces must not be conflated. The native ring successor T is a step in canonical address order. The geometric word rewrite is a change of chain conformation. A measurement/readout R extracts a specified property of the state. In particular, T does not become a protein move simply because it is a reversible permutation.

The twisted one-inverter clock and temporal reader

The "One Inverter" paper derives the affine twisted rotation $\tau(x_0 \ldots x_{n-1}) = x_1 \ldots x_{n-1} \neg x_0$, so after k ticks the first k departing bits are returned complemented. At k=n it is global complement; at k=2n the state returns. The inward terminal reader is exactly the XOR of the two central word bits. Under τ, its temporal trace reads each adjacent difference around the cyclic word, with one known inverted crossing.

$\tau^k(x) = \mathrm{rotl}_k(x) \oplus (2^k - 1)$, $0 \leq k \leq n$; $\tau^n = C$, $\tau^{2n} = I$.

$r(x) = x_{n/2-1} \oplus x_{n/2}$; $\bigoplus_{t=0}^{n-1} r(\tau^t x) = 1$. [even n]

The n−1 independent adjacent differences plus one native face bit recover the word. The two-bit closed complement-feedback rule is τ^2, not a new primitive. None of this creates a physical molecule; it gives a provably executable, invertible substrate for how a local reader can obtain and return distinctions.

Odd-lane locks and relational deviations

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For odd q and n=qs, the alternating-complement lift $\Lambda q(u)=u\|Cu\|u\|\ldots\|u$ is invariant under the twisted clock and intertwines J, C, and P. A center trace of period s identifies this lock and supports exact reconstruction through the first s-bit lane. In the full q-lane state, de-staggered lane differences and one anchor preserve all n bits.

$\tau_n \Lambda q = \Lambda q \tau_s$; $P_n \Lambda q = \Lambda q P_s$; $dH(\Lambda q\, u, \Lambda q\, v) = q \cdot dH(u,v)$.

These differences were formerly called "defects." They are lawful relational deviations from a selected lock—not failures of the transformation. Under common affine transport, differences rotate and preserve weight. An external input can alter them at a known boundary, but selecting an input law is not already derived by the native permutation.

The native address T, the twisted clock τ, and the geometric pivot defined later remain separate typed operators. The exact E/G route has n bits for a general n-bit input. A locked multi-lane family instead requires only its independent local state and the stated lift rule; the reduction comes from a demonstrated invariant restriction, not from labeling an arbitrary large input "locked."

Which operations the embedding preserves

The One Inverter temporal read is an operational return path, not just a yes/no label. For $n=2h$, define the cyclic adjacent differences $D_i=x_i \oplus x_{i+1}$ (indices modulo n) and the observed center trace $y_t=r(\tau^t x)$. The twisted boundary contributes exactly one inverted crossing. Over n ticks the center presents n relations; their XOR equals one, and $n-1$ of them together with the retained canonical face bit reconstruct the entire original word. The E/G inward/outward route supplies a separate exact return by retained qualifiers. Neither return is compression of arbitrary n-bit words.

$y_t = D_{(h-1+t)} \bmod n \oplus [t=h]$, $\quad \bigoplus_{t=0}^{n-1} y_t = 1$; $\quad (y_0,\ldots,y_{n-2}, \text{face}) \leftrightarrow x$.

The two masks must not be conflated. The mirror shell lamps are $g_j=x_j \oplus x_{n-1-j}$, the paired entries in $x \oplus J(x)$. The distinct partner-fold mask is $x \oplus P(x)$. They agree away from palindromes, but on a palindrome $x \oplus J(x)=0$ while $x \oplus P(x)=1^n$. Therefore a statement that identifies all shell lamps with the partner-fold mask is false without the palindrome qualification.

The odd-lane lift $\Lambda q(u)=u\|Cu\|u\|\cdots\|u$ for odd q intertwines τ, τ^{-1}, reversal J, complement C, and the conditional native partner P. It also preserves the canonical face and center, and preserves the order of lifted canonical representatives. It does NOT intertwine the native ring successor T: in general $T_{(qs)}\Lambda q \neq \Lambda q T_s$. Global address gaps between successive lifted representatives are non-unit. This is the precise boundary for transplanting any local operator to a wide register.

$\tau_{(qs)}\Lambda q = \Lambda q \tau_s$; $J_{(qs)}\Lambda q = \Lambda q J_s$; $C_{(qs)}\Lambda q = \Lambda q C_s$; $P_{(qs)}\Lambda q = \Lambda q P_s$; $T_{(qs)}\Lambda q \neq \Lambda q T_s$ (generally).

Accordingly, a Hamiltonian assembled from $T+T^\dagger$ does not inherit a molecular or synchronized-lane kinetic interpretation from the τ-lift theorem. The kinetic edge map must be supplied independently from the molecularly admissible transition graph. At the register level, the exterior two-bit input is already a typed injection: deviation from closed complement feedback is $\varepsilon=b \oplus (a \oplus 3)$, carried by the known shift interface. The remaining OPEN link is what physical interaction supplies b and with what rates or amplitudes.

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Compatibility spheres and reader quotient

Compatible partial readings of n bit positions form the boundary of an n-dimensional cross-polytope, homeomorphic to S^{n-1}. Global complement is antipodal, producing RP^{n-1} on quotient. The lamp reader makes a still coarser identification. These results describe a topology of partial information; a physical metric, chemical coordinates, energy, and force law are not supplied by the topology. Native P also fails to preserve Hamming adjacency precisely across the palindrome/non-palindrome boundary.

The topology of partial readings is relevant to interface design because it makes explicit what distinctions a reader identifies. It must not be substituted for Euclidean molecule coordinates or a kinetic metric. Likewise, a mirror quotient and the coarser shell-lamp quotient are different equivalence relations.

3. Quantum representation, operator contracts, and scope

A reversible finite map F induces a unitary permutation on the Hilbert basis, $U_F|x\rangle=|F(x)\rangle$. This is a mathematical bridge from native compilers to quantum circuits. On its own, however, a basis permutation only relabels amplitudes; starting in one basis state it does not create a superposition. Relative phases and interference enter when additional quantum operations are specified. The supplied runs use a Hermitian generator H=V+K, with diagonal contact energies V and coherent hopping K.

$U_F|x\rangle = |F(x)\rangle$; $H = V + K$; $U(t)=\exp(-i H t)$ ($\hbar=1$).

For a cyclic computational translation T_N, $K=-\gamma(T_N+T_N^\dagger)$ is a quantum walk. Its Fourier eigenvalues are $-2\gamma\cos(2\pi k/N)$; hence a QFT efficiently diagonalizes that particular K. K does not execute the exact discrete successor T_N: its exponent generates coherent mixing. A Hermitian logarithm of T_N can execute the exact successor at a chosen time, but that generator has phase-branch ambiguity and no established molecular locality. These are distinct typed operations.

Let $X_m = \{0,1\}^m$ denote turn words, $V_m \subseteq X_m$ the legal self-avoiding words, and $\mathcal{H}_m = \mathbb{C}^{V_m}$ the associated complex vector space. A reversible state transformation is a bijection F on its specified domain. It induces the permutation unitary $U_F|x\rangle = |F(x)\rangle$. This is a representation theorem, not an assertion that matter physically follows F.

$x \in V_m \quad \rightarrow \quad |x\rangle \in \mathcal{H}_m, \qquad U_F|x\rangle = |F(x)\rangle$

There are three independent tests for any claimed quantum-compatible compiler. First, the state map must be well typed and invertible or be embedded with its missing information retained. Second, the operator must preserve the intended domain. Third, the reduced representation must reproduce the chosen evolution and observables, not merely the same static labels. Unitary representability establishes the first point for a permutation. A Hermitian Hamiltonian and a quantitative propagation audit address the latter two, for the specified finite model only.

$U_F|x\rangle = |F(x)\rangle \quad \cdot \quad [H, P_{valid}] = 0 \quad \cdot \quad E\, H_{phys} = H_{comp}\, E$ (when physically justified)

PART II / WHY THE FIRST PROTEIN HAMILTONIAN WAS NOT YET A MOLECULAR COMPILER

4. The initial four-state HPPH demonstration

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The first executable bundle fixes the initial bond east and encodes two turns as a two-qubit word. Its four square-lattice configurations have contact energies [−1, 0, 0, −1]. In native width-two ring order (00,10,01,11), it uses γ=0.7 and an assumed cyclic hopping mixer. At dimensionless time t=1 the supplied direct exponential gives a contact probability 0.7717491215, up from 0.5. This is a coherent redistribution between states, not relaxation to a unique fold. The amplitude probability oscillates in time; no solvent or dissipation is included.

This open four-residue example is retained because it separates an elementary quantum-walk computation from a physical folding claim. A rise in favorable-contact probability at one time can be reproduced by coherent mixing with a chosen kinetic operator. In a closed Hermitian model it neither demonstrates irreversible relaxation nor identifies the environmental coupling responsible for real folding.

5. The six-residue cyclic PHPPHP domain and its eight-address embedding

The second model closes the six-residue backbone on a 2D square lattice, keeps all residue positions distinct, and fixes the first bond east. It produces exactly six valid conformations, all rectangles of perimeter six; the supplied HP contact energies are [−1,−1,0,0,0,0]. Three qubits give eight addresses, so the two additional states (native addresses 3 and 7) are declared invalid and given potential +10. This is an encoding choice; the two register padding states are not alternate molecular geometries.

Native address	Native R_3 word	Catalog index	Potential	Status
0	000	0	−1	valid; contact
1	001	1	−1	valid; contact
2	010	2	0	valid
3	110	6	+10	invalid padding
4	011	3	0	valid
5	101	5	0	valid
6	100	4	0	valid
7	111	7	+10	invalid padding

Original native R_3 address-to-conformation catalog. Invalid rows are computational padding, not alternative molecule shapes.

The ring-address mapping is exact as a catalog index. It is not a derivation of which conformational changes are kinetically or chemically possible. Neither a loop in the residue backbone nor a loop in the computational addresses entails the other.

Geometry-versus-address audit

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I compared all 15 unordered pairs of the six coordinate lists supplied in cyclic_report.json, with residue identity and anchored first bond preserved. Exactly six pairs differ in three residue positions and nine pairs differ in four; no pair differs in one or two positions. Mirror symmetry pairs catalog indices (0,1), (2,5), and (3,4), preserving their assigned HP energy. These are exhaustive finite facts of the declared six-state lattice model.

A one-site coordinate-update grammar would therefore give zero edges inside this catalog, so nontrivial rearrangement requires a different grammar (concerted moves, a richer intermediate state domain, or continuous 3D dynamics). This does not prove a three-residue physical move is realizable; it only bounds how many stored lattice coordinates differ between existing entries.

Geometry result	Exact value	What it determines
Valid conformations	6	Configuration domain in this toy model
Pairs differing in 1 or 2 residue positions	0 of 15	No single-site/dual-site catalog edge
Pairs differing at 3 positions	6 of 15	Smallest discrete catalog displacement
Pairs differing at 4 positions	9 of 15	Remaining catalog displacement
Mirror-equivalent pairs	3	Symmetry of the potential, not a move law

Exact pairwise geometry distinctions among the six catalog conformations.

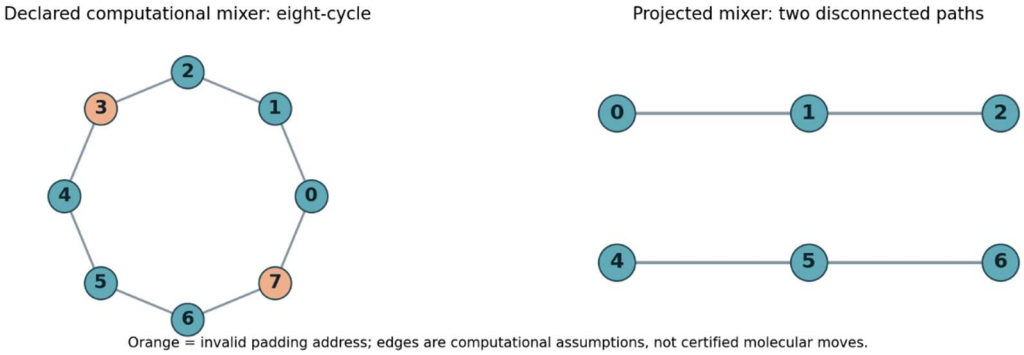


Figure 2. Original eight-address ring and its exact valid-only projection. The two invalid sites connect otherwise separated conformational catalogs.

6. Domain leakage under assumed cyclic hopping

Let P project onto native addresses {0,1,2,4,5,6} and Q=I−P onto the padding addresses {3,7}. The assumed eight-cycle kinetic $K=-0.7(T_8+T_8^{\dagger})$ has four valid/invalid edges: (2,3), (3,4), (6,7), and (7,0). Therefore $QKP \neq 0$. A diagonal padding penalty cannot remove these edges.

$[H,P] = 0 \iff QHP = 0$ [H Hermitian].

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For the supplied initial state $|\psi_0\rangle$ uniform on the six valid addresses, $Q|\psi_0\rangle=0$. Expand the continuous-time amplitude: $Qe^{(-iHt)}|\psi_0\rangle = -it\,QK|\psi_0\rangle + O(t^2)$. Hence the invalid-state probability begins at $49t^2/75 + O(t^3)$, independently of any finite penalty. The measured coefficient is 0.653333333333 and the operator coupling norm $||QKP||_2 = 0.98994493661$.

$p_{invalid}(t) = ||Qe^{(-iHt)}\psi_0||^2 = (49/75)t^2 + O(t^3)$.

The projector criterion answers an admissibility question that energy penalization does not. If $QHP \neq 0$, the amplitude has a first-order path into the invalid sector even if the invalid basis vectors start unoccupied. Because the transition amplitude is $O(t)$, its probability begins at $O(t^2)$. The diagonal penalty does not enter that leading coupling.

The original model suppresses departures with $+\Lambda$, but coherent amplitudes still enter the invalid states. Exact propagation at t=1 produces the following values. Setting K_valid=PKP creates a truly invariant valid sector for any diagonal V, giving exactly zero padding leakage without requiring $\Lambda\to\infty$.

Penalty Λ	Eight-cycle leakage at t=1	Projected K_valid leakage
0	0.324402465	0
10	0.006480472	0
50	0.000151516	0
100	0.000023742	0

Penalty versus leakage in exact eight-cycle evolution (dimensionless t = 1).

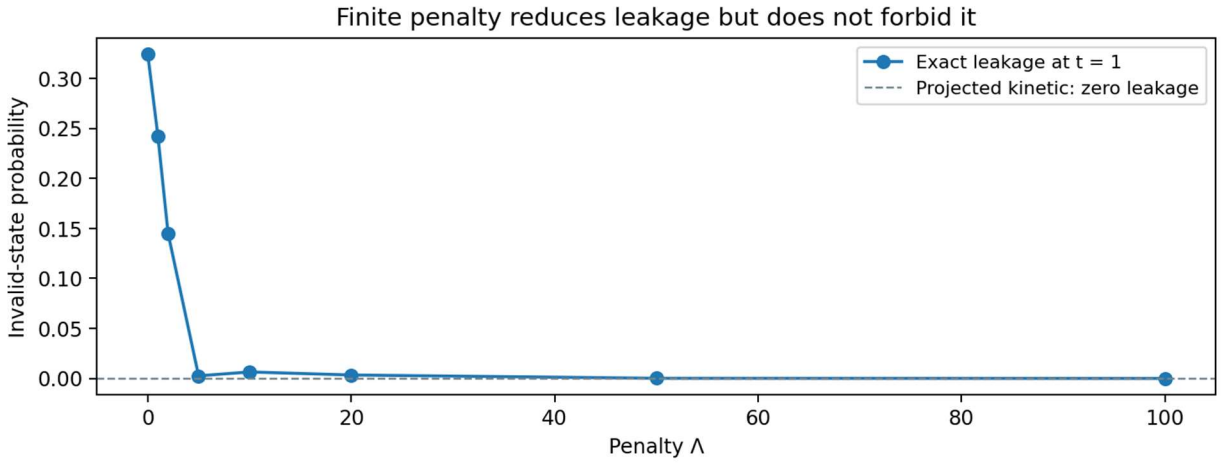


Figure 3. Finite penalty suppresses occupation at selected times; the exact projected operator has no padding subspace to occupy.

Projection exposes a second issue. The valid-only graph is 0—1—2 and 4—5—6, two disconnected paths. Contacts occur only at 0 and 1, both in the first component. The initial uniform six-state wavefunction places

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exactly 1/2 of its probability in the other component, and K_valid can never transfer that portion into either contact state. Therefore contact probability is at most 1/2 under this projected mixer, for this initial condition. At t=1 it is 0.463631173441 with zero invalid leakage. This is a connectivity obstruction, independent of padding penalties.

$p_{contact}(t) \le 1/2$ for K_valid and the specified $|\psi_0\rangle$.

7. Virtual bridges from the invalid sector

A finite-energy elimination of padding is not identical to deletion. Partition H into valid P and invalid Q blocks. For an eigenvalue parameter E away from the invalid block spectrum, the Schur complement is exact:

$H_{eff}(E) = H_{PP} + H_{PQ} (E - H_{QQ})^{-1} H_{QP}$.

Here $H_{QQ}=\Lambda I_2$, since the two invalid sites are not adjacent in the eight-cycle. $H_{PQ} H_{QP}$ contributes 0.49 to the diagonals of valid addresses 0,2,4,6, and 0.49 to each off-diagonal pair (0,6) and (2,4). Thus the penalty model retains energy-dependent virtual bridges joining the two paths, with leading amplitude approximately $-0.49/\Lambda$ for large Λ and E small compared with Λ. The projected model deletes both the forbidden population and those virtual connections.

Those bridges are not evidence for protein dynamics. They follow from the originally assumed computational edges through invalid padding states. What appears to be a new available folding route can be an artifact of the address graph. This explains why high-penalty low-energy spectra can differ from the spectrum obtained by deleting padding.

This exact Schur-complement result explains why a large-penalty model and a projected valid-state model can disagree even at low energy. The virtual bridges are generated by the forbidden computational edges, not by demonstrated protein transitions. Removing a mathematical padding state may therefore remove a dynamically influential modeling artifact; its previous influence is not evidence that the absent configuration was physically traversed.

8. Product-formula accuracy and Fourier implementation

For $H=V+K$, the Strang product $S(\delta)=\exp(-iV\delta/2)\exp(-iK\delta)\exp(-iV\delta/2)$ is a second-order approximation to $\exp(-iH\delta)$. The supplied nested commutators for $\Lambda=10$ reproduce independently: $\|[V,K]\|_2=10.406248$, $\|[V,[V,K]]\|_2=109.882164$, and $\|[K,[K,V]]\|_2=24.111397$. Their reported bound coefficient is 6.587707, giving an $O(t^3/r^2)$ error bound, not an exact measured error.

Strang steps	Measured operator error at t=1	Bound $6.587707/r^2$
4	0.082770330	0.411732
8	0.018193978	0.102933
16	0.004416348	0.025733
32	0.001096147	0.006433

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Strang steps	Measured operator error at t=1	Bound $6.587707/r^2$
64	0.000273545	0.001608
128	0.000068356	0.000402

Strang error in the specified eight-state Hamiltonian; bounds and measured operator errors are distinct quantities.

The second-order scheme is accurate for this chosen H; its numerical convergence does not validate H as a molecular Hamiltonian. The earlier source report, not re-derived in the present §147 audit, counts one unoptimized three-qubit split step as 34 CNOT, 32 Rz and 6 Hadamards after decomposing controlled-phase and SWAP operations. QFT diagonalizes the cyclic kinetic term, but it generally will not diagonalize a geometry-derived graph that loses translation symmetry. Gate-cost claims also depend on synthesis, compilation, connectivity and error tolerance.

These commutators quantify numerical approximation cost for the chosen matrix H. They do not validate its kinetic edges. Large penalty jumps increase nested-commutator magnitudes even as they reduce invalid population, an example of an algorithmic cost incurred by repairing a modeling choice through energy scale alone.

9. The "way home" in an enlarged quantum system

A closed unitary U preserves inner products. If six distinct orthogonal input conformations all appear as the same pure protein output $|g\rangle$ under a larger unitary, their lost distinctions cannot vanish. For a common initial environment $|o\rangle$, write $U|i\rangle|o\rangle = |g\rangle|e_i\rangle$. Unitarity implies $\langle e_i|e_j\rangle=\delta_{ij}$. The environment must distinguish the six origins even though the protein reader sees one state.

$\langle i|j\rangle = \delta_{ij} \Rightarrow \langle e_i|e_j\rangle = \delta_{ij}$ if $U|i,o\rangle = |g,e_i\rangle$.

A finite demonstration is SWAP: $|i\rangle_{protein}|o\rangle_{env} \rightarrow |o\rangle_{protein}|i\rangle_{env}$. All six protein outputs look identical while the full route is invertible; the original label remains in the environment. This construction does NOT model heat, solvents, reaction rates or folding; it proves what kind of hidden return channel an apparently many-to-one protein evolution requires. The classical analog is retaining the departing two-bit pair when an external port inserts an independent incoming pair.

For real folding, reduced protein dynamics may be nonunitary because the protein exchanges energy and information with solvent, vibrations, chaperones, and other degrees of freedom. Any assertion that the ring explains folding must identify this coupling rather than treating an amplitude oscillation in an isolated toy Hamiltonian as a dissipative biological fold.

The return record distinguishes reversible total-state evolution from apparently many-to-one output at a restricted reader. The toy SWAP construction is a proof about information preservation, not a heat-bath simulation. It defines a contract for later protein–solvent models: any apparent convergence of many initial states to one observed state must be accompanied by distinguishable environmental states or an explicitly nonunitary effective description.

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PART III / REPLACING CATALOG HOPPING WITH THE HONEYCOMB ROUTE COMPILER

10. Honeycomb turns create geometric states directly

Fix the first chain bond eastward. Each subsequent turn $b_i \in \{0,1\}$ rotates the heading by +60° or −60°, respectively, using one of six honeycomb directions. An m-bit word therefore gives m+2 lattice sites: the two sites of the initial bond followed by m new sites. A state is legal precisely when these m+2 coordinates are pairwise distinct. The geometric decoder is injective once the first bond and turn convention are fixed, because successive headings recover the input turns.

$V_m = \{x \in \{0,1\}^m : |\{r_0(x), \ldots, r_{m+1}(x)\}| = m+2\}$

For the HP sequence s, define $E_{HP}(x)$ as minus the number of nonconsecutive nearest-neighbor H–H lattice contacts. This is a deliberately coarse contact potential. It has neither a solvent free-energy model nor continuous torsion dynamics.

The domain shift is decisive. Rather than number six pre-enumerated geometries and then guess an eight-cycle mixer, the compiler now builds each chain from its turn word. A legal word is accepted by the geometric self-avoidance constraint. Molecular sequence enters separately through the contact-energy read. The turn map has a direct inverse because the sequence of headings reconstructs the binary turns once the initial frame is fixed.

The first bounded-support geometric rewrite

The shortest nontrivial substitution preserving both the substring endpoint and its outgoing heading is the four-turn exchange 1001 ↔ 0110. It moves two internal residues around a single hexagonal face while leaving the remainder of the chain fixed. Applying the identical mask twice returns the original word; a proposed edge is retained only if both endpoints are legal. No padding or large penalty is required.

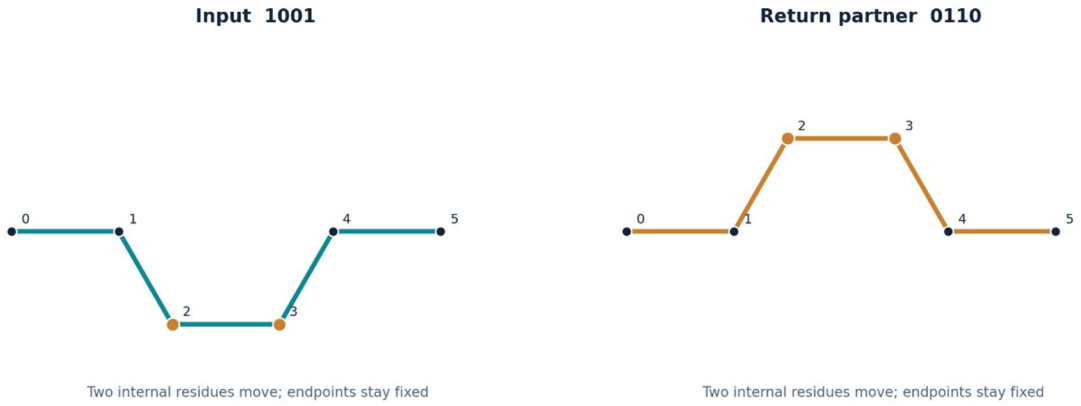


Figure 4. Local four-turn hexagon exchange 1001 ↔ 0110. The inner residues move; the two ends and outside chain remain fixed.

By contrast, flipping one interior turn bit rigidly rotates the downstream tail. That is reversible and can be legal on the lattice, but its spatial support grows with chain length. The distinction between a legal move and a bounded-support move is a central result of the §146 audit [2].

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A state-space edge now has a geometric certificate: applying the defined operation changes only specified coordinates and returns a self-avoiding chain. An abstract reversible pivot remains legal but need not be physically local. Locality and reversible accessibility will be reported independently throughout the rest of the paper.

11. Complement reflection and the exact two-sheet quantum split

Define C(x) = x ⊕ (2^m−1). In the honeycomb turn encoding C reflects the chain geometry and has no fixed binary words. Self-avoidance and HP contact counts are invariant under this reflection for any fixed residue sequence. Let A be the adjacency matrix of a declared C-stable legal-move graph and let V be the diagonal HP potential. With a freely specified hopping strength γ, define

$H = V - \gamma A, \quad U_C |x\rangle = |Cx\rangle, \quad [H, U_C] = 0.$

The commutator follows by checking V(Cx)=V(x) and that every allowed edge x↔y has partner edge Cx↔Cy. Both requirements were explicitly audited for the finite domains. The model therefore separates into the eigenvalue +1 and −1 sectors of U_C, whether or not the legal graph is connected.

$|a,+\rangle = (|a\rangle + |Ca\rangle)/\sqrt{2}; \quad |a,-\rangle = (|a\rangle - |Ca\rangle)/\sqrt{2}.$

Signed quotient and parity basis

Pick one representative a from each pair {a,Ca}. Let A_0 contain representative-to-representative moves and A_1 contain representative-to-complemented moves. Self-loops and parallel same/opposite-face edges must be retained as distinct matrix contributions. An orthogonal two-point Fourier transform on the face bit then gives the exact decomposition

$H_+ = V_q - \gamma(A_0 + A_1), \quad H_- = V_q - \gamma(A_0 - A_1).$

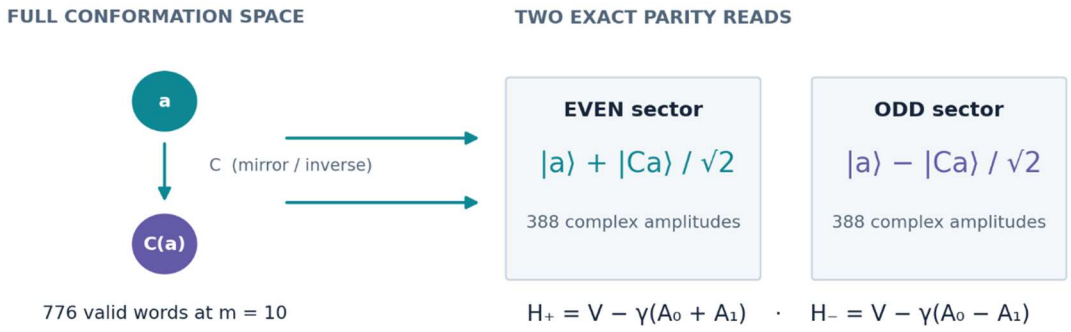


Figure 5. Mirror-pair quotient with its retained face bit. A two-point Fourier transform gives the even and odd Hilbert sectors.

The sign of A_1 is not an adjustable penalty. It is the character of the complement operation in the odd representation. For a mirror-even state, the outward return map copies each reduced amplitude equally to a and Ca; for a mirror-odd state, it copies with opposite signs. For a general state, both sectors are required.

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For any C-equivariant adjacency and C-invariant potential, the operator commutation is exact algebra: the sector decomposition follows without requiring spatial connectivity. A general state has independent amplitudes in both sectors. One-sector execution is exact only for a prepared state in that sector, or for an observable question that is provably answerable there.

$\psi_a = (\phi_+(a) + \phi_-(a))/\sqrt{2}$; $\psi_{Ca} = (\phi_+(a) - \phi_-(a))/\sqrt{2}$.

12. Holonomy is conditional on the move vocabulary

A closed route in the mirror quotient lifts either to a closed route on the original sheet or to its complement. The latter carries product of sheet-transition signs −1, a nontrivial Z_2 holonomy. Its existence in a quotient component is equivalent to the original connected component containing both x and Cx; it does not require the entire graph to be connected. A global claim about "the" holonomy must therefore state the admissible move set.

The local hexagon mask toggles four consecutive bits; its m−3 masks are linearly independent over GF(2), leaving codimension three. Adding the final-turn flip leaves codimension two. The complement mask is representable by these moves only when $m \equiv 0 \pmod 4$, or with the terminal bit included when $m \equiv 0$ or $1 \pmod 4$ [2]. This includes a proven all-length *legal* construction, not only finite checks: build the route from repeated blocks 1001 and replace each by 0110; the intermediate headings remain in {−60°,0,+60°}, so the chain has strictly increasing longitudinal coordinate and cannot self-intersect. The optional final flip covers $m \equiv 1 \pmod 4$. These are sufficiency constructions for those move vocabularies, not a claim that the full local graph becomes connected.

Claude's independent §146 audit correctly sharpened the interpretation of that algebraic deficiency: a low-dimensional GF(2) quotient does not count geometric connected components. At m=10 the hexagon-plus-terminal graph has 192 components, far more than the four classes suggested by two missing binary directions. Frozen residue coordinates and self-avoidance create additional obstructions. An extension must address legality and reachable geometry, not just binary rank.

The sheet transition σ along quotient edges is retained with multiplicity: replacing a signed multigraph with an unweighted simple graph can erase opposite-face paths or double-count a self-loop. A loop returning to the opposite sheet has product $\sigma = -1$. Such a sign is a property of the chosen graph covering, not by itself a Berry phase or a measured physical flux.

13. Exact connectivity and energy reachability

The audited ten-turn domain has 776 legal conformations of the 12-residue sequence HHPHPPHHPPHP, including 32 states with at least one H–H contact. Graph components were counted by exact word-edge enumeration. "Can reach contact" means connectivity in the defined graph, not a finite-time transition probability.

Move vocabulary	Edges	Components	Reach a contact
All legal one-bit pivots	3,352	1	776 / 776

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Move vocabulary	Edges	Components	Reach a contact
Pivots restricted to ≤4-residue tail	1,374	58	88 / 776
Only local hexagon rewrites	396	404	42 / 776
Hexagon rewrites + terminal turn	768	192	44 / 776
Hexagon rewrites + one-/two-residue moves at both ends	1,840	14	612 / 776

Finite m = 10 honeycomb graph census with 776 valid conformations, 32 contact configurations, and an independently reproduced both-end-move extension from §146.



Figure 6. Five legal move vocabularies on the same 776-state domain. The physically equivalent front-end moves require undoing the arbitrary fixed-first-bond reference orientation; local geometric support and graph reachability are distinct.

The broad structural limitation is not a surprise to polymer theory. Madras and Sokal established nonergodicity results for finite repertoires of local, length-conserving self-avoiding-walk algorithms [3]. Their theorem should not be silently transferred to the present honeycomb move vocabulary without matching its hypotheses, but it gives an essential negative control against claiming that adding one or two local masks makes all conformations mutually reachable. Pivot methods and bilocal moves also have established histories [4].

The 404 components under local hexagon rewrites do not signal an error in the move compiler; they record the reachable subsets of that vocabulary. With an end move at only the terminal side, 192 components remain. The fixed first bond is a reference-frame choice, not a prohibition against motion at that end. The independent §146 follow-up identifies one- and two-residue end motions at both chain ends: the corresponding 1,840-edge graph has 14 components, and 612 of 776 conformations can reach at least one contact state. This substantially improves but does not restore ergodicity. Its addition does not alter the known local-move obstruction or supply physical rates.

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13.1. A precondition gate, an isolated zigzag family, and the next local return

The four-turn 1001↔0110 move is conditional, not an unconditional XOR direction. It can fire only when the local four-turn window is 1001 or 0110. For even $m=2h\geq 4$, write the interior pair-parity vector $p(x)=(b_2\oplus b_3, b_4\oplus b_5, \ldots, b_{m-4}\oplus b_{m-3})$. The four permitted one- and two-residue end moves leave these interior pairs unchanged. Even-aligned hexagon rewrites preserve each pair parity. An odd-aligned hexagon rewrite requires its central pair parity to vanish; hence it cannot fire when every interior pair has parity one. These are precondition constraints, not missing entries in a numerical data table.

$Z_m=\{x\in V_m: p(x)=(1,\ldots,1)\}$; $|Z_m|=2^{((m+4)/2)}$ (even m, counting allowed bit assignments).

For even $m=4,6,8,10,12,14$, exhaustive geometry showed that every word in Z_m is self-avoiding and the restricted both-end/hexagon graph has Z_m as exactly one closed component. Its cardinalities were 16, 32, 64, 128, 256, 512. The geometric interpretation is a zigzag trap: no operation in that vocabulary can supply the required change of an interior pair without violating its own precondition. The all-width word-count formula is algebraic; the complete self-avoidance and one-component claims beyond the tested range are not inferred from a finite plot.

This exposes the next missing WHAT. A nontrivial rewrite changing one isolated internal residue would create a four-edge cycle in the honeycomb graph, which does not exist. Two changing adjacent internal residues necessarily exchange opposite three-edge arcs of one hexagon: they are already covered by 1001↔0110. Three changing internal residues would require an independent eight-edge cycle; the honeycomb has none. The first geometrically distinct fixed-exterior return is therefore an exchange of two five-edge arcs on the ten-edge boundary surrounding two adjacent hexagonal faces. It moves exactly four internal residues and leaves both endpoints—and every residue outside the window—fixed. The old five-edge arc is its explicit inverse qualifier.

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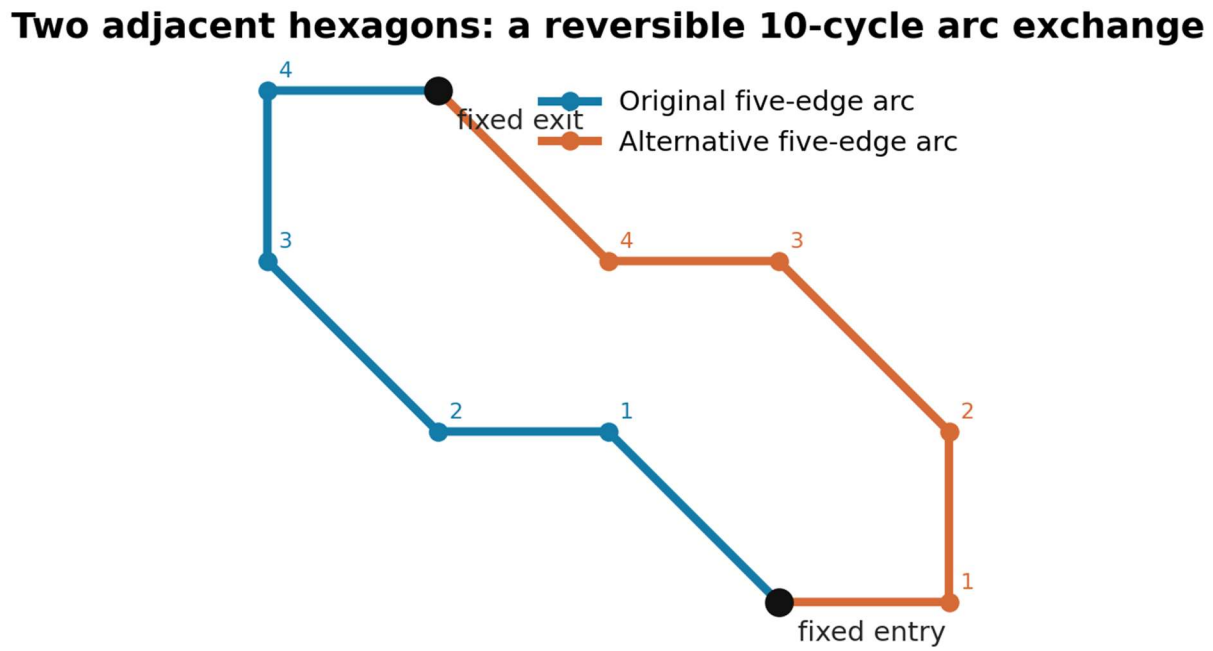


Figure 9. Two five-edge routes between the same fixed endpoints enclose two neighboring honeycomb hexagons. Replacing one arc by the other moves exactly four interior residues and retains a literal inverse route. These are discrete lattice configurations; continuous collision-free dynamics remains untested.

The ten-cycle edges were constructed from endpoint-compatible lattice paths, not from native successor addresses. A candidate edge was retained only when the entire resulting word decoded, all outside residue coordinates were unchanged, the replacement had four changed internal sites, and the resulting conformation was self-avoiding. Every retained move is a reversible discrete geometric rewrite. The mirror C carries this edge set into itself exactly in all tested widths.

The resulting finite graph computations are below. The Z_m-exit column counts undirected ten-cycle edges crossing between the precondition-locked zigzag family and the rest of the legal conformation domain; it is not a sample of energetic transitions.

Turn bits m	**Legal states**	**Both-end components → with 10-cycle**	**10-cycle edges**	**Z_m-exit edges**
6	58	2 → 1	12	8
8	216	5 → 3	78	16
10	776	14 → 3	400	32
12	2,796	35 → 5	1,882	64

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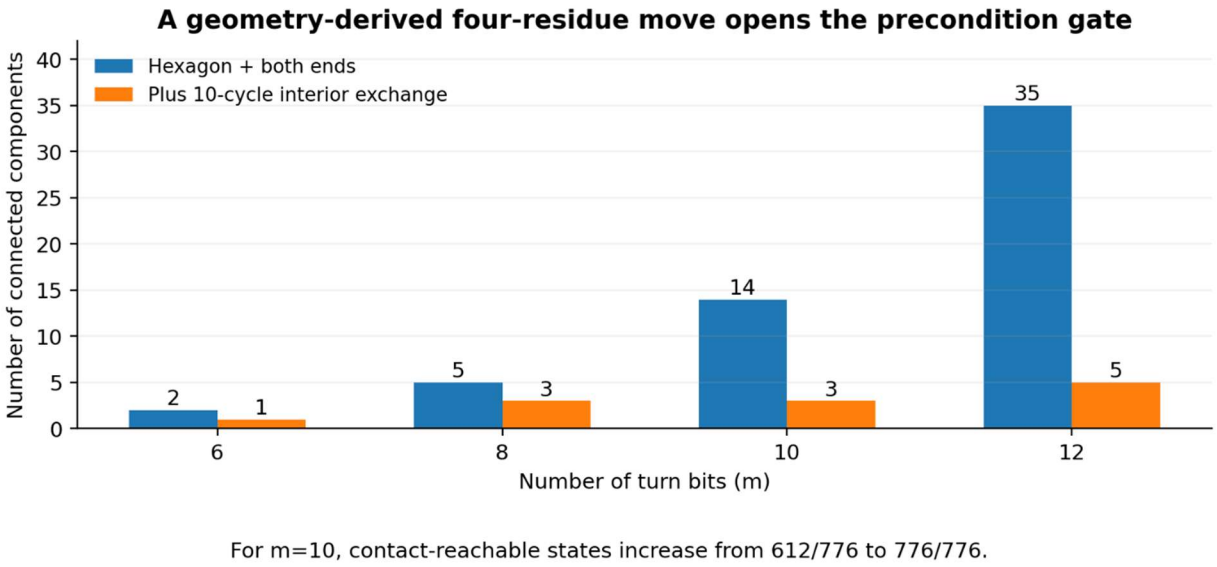


Figure 10. Connectivity after adding the first distinct bounded-support interior return. Both-end/hexagon dynamics alone traps the zigzag family. The ten-cycle exchange crosses its gate at every executed width m=6,8,10,12.

For the fixed m=10 HP chain HHPHPPHHPPHP, the earlier local vocabulary left 612 of 776 states in components containing a hydrophobic contact. Adding the 400 distinct ten-cycle transitions produces a 2,240-edge graph with only three components (largest 766 states), and every one of the 776 legal conformations can now reach at least one of the 32 contact states. The formerly isolated zigzag family is no longer a separate component. This does NOT prove global ergodicity: three components remain, and it does NOT assign chemical rates to these proposed four-residue exchanges.

An explicit outward route from the previously trapped benchmark input is: 0000100010 → 0100100010 → 0011000010 → 0111000010 → 1100010010 → 1000010010. The fourth transition is the new ten-cycle exchange; the final state has HP contact energy −1. Every intermediate state is self-avoiding. Reversing those five recorded transitions returns the original input exactly. A second seven-step route takes the zigzag word 0101010101 to the contact word 0111100101, using two ten-cycle exchanges. These are finite graph paths, not claims of real-time protein kinetics.

The resulting OPEN physics is sharply typed: whether the ten-cycle exchange has a continuous sterically feasible realization, what activation barrier or solvent-assisted pathway enables it, and what rate or quantum coupling should be assigned. The demonstration is an admissibility-and-reachability result in a two-dimensional honeycomb HP model, not a mechanistic prediction for a three-dimensional protein.

PART IV / ACTUAL QUANTUM-STYLE COMPUTATION ON ORDINARY HARDWARE

14. Independent full-space and parity-reduced propagation

The new independent sparse-matrix audit [9] reconstructs V_m from integer-coordinate geometry, builds A from legal transitions, constructs H, and independently assembles H_+ in the quotient representation. It

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prepares a complement-even pair $\psi_0=(|x\rangle+|Cx\rangle)/\sqrt{2}$ and propagates the same dimensionless Hamiltonian both ways:

$\psi(t) = \exp(-i\,H\,t)\,\psi_0$; $\phi_+(t) = \exp(-i\,H_+\,t)\,\phi_+(0)$.

The outward reconstruction copies $\phi_+(t)/\sqrt{2}$ to both members of each mirror pair. The §146 source audit tested t = 0, 0.2, 0.5, 1, and 2 for m = 8 hexagon and m = 10 hexagon, hybrid, and unrestricted-pivot Hamiltonians. Full-space and reduced-sector evolution agree to machine precision with SciPy sparse expm_multiply (largest discrepancy $\leq 4.4 \times 10^{-16}$ in the five-time verification); norm was preserved within 5×10^{-12}. Claude's independent §147 eigendecomposition returned a larger but still rounding-scale discrepancy of 1.7×10^{-15}, failing its previously sealed numerical threshold of 5×10^{-16}. The unitary block equivalence is algebraically exact, but a fixed floating-point tolerance is solver dependent. The m = 8 test starts from the mirror-even pair at 00001000; m = 10 uses 0000100010. All propagation ran on conventional hardware.

m, legal operator	Full dimension	Even dimension	Largest |Δ amplitude|
8, hexagon	216	108	Machine precision (expm_multiply)
10, hexagon	776	388	Machine precision (expm_multiply)
10, hexagon + end	776	388	Machine precision (expm_multiply)
10, all pivots	776	388	Machine precision (expm_multiply)

Full and reduced state dimensions; numerical return error for all four audited operator configurations.

Energy contact observable under two lawful graphs

For the 12-residue sequence HHPHPPHHPPHP, the mirror-even source at turn word 0000100010 has no H–H contact initially. Under unrestricted legal pivots its classically propagated contact probability is 0.0017113 at t = 0.5, 0.0430001 at t = 1, and 0.0335927 at t = 2. These are three values along an oscillatory curve, not evidence of a peak at t = 1: a dense 2,001-point time read over $0 \leq t \leq 10$ places the first local peak near t = 1.405 and the largest sampled value, ≈ 0.101858, near t = 6.255. Under hexagon-plus-terminal moves the *same source* belongs to a component of exactly two words, 0000100010 and 0000100011. Neither has a hydrophobic contact, so its contact probability is identically zero for every t by support; that special two-state island does not characterize local moves in general. The rates and amplitudes still use the model choice γ = 0.7 and dimensionless time.

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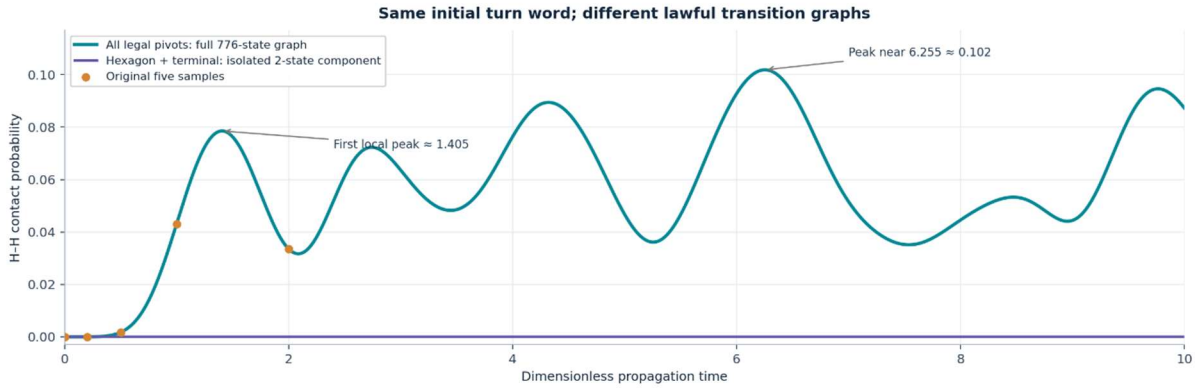


Figure 7. Dense 0–10 classical propagation of the same mirror-even source in two declared Hamiltonians. Orange markers identify the original five time samples. The local trace is identically zero because its source occupies a two-state, contact-free component; it is not a general null result for locality.

The comparison has two separate logical controls. The full 776-component propagation supplies a numerical reference for the separately assembled 388-component parity block. The local two-state island versus the connected unrestricted-pivot graph then demonstrates how one specified initial state can have different observable contact access under different move vocabularies; it does not establish that every local component is contact-free. A correct compiler must not confuse representation equivalence with physical equivalence of two different Hamiltonians.

15. Resource accounting and the low-power thesis

A model with N legal conformations needs N complex amplitudes for its full pure-state representation. Because complement acts freely, each parity sector has N/2 coordinates. At m=10 this is 776 versus 388 complex128 amplitudes, or 12,416 versus 6,208 bytes of raw state-vector storage. For a dense Hamiltonian one block occupies one-quarter of the full matrix entries; this is a storage comparison only and does not measure wall-clock speed. The actual experiments use sparse operators whose memory costs depend on edges and implementation details.

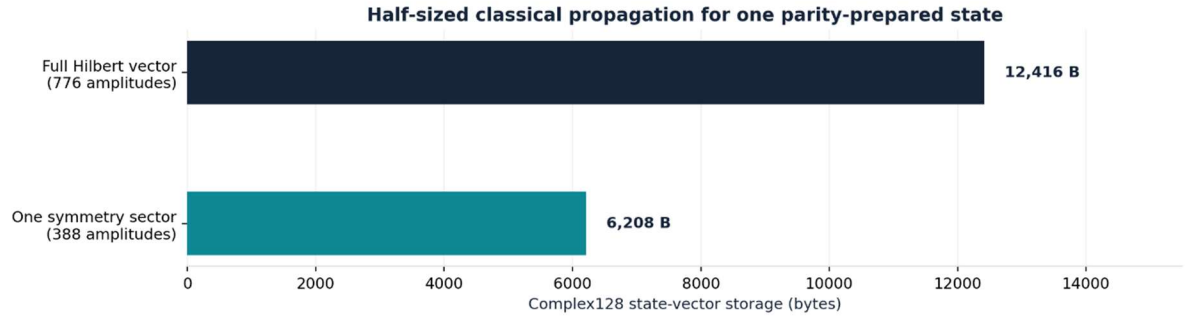


Figure 8. Raw complex128 state-vector storage for the ten-turn example; a parity-selected state requires half the full vector.

A mirror symmetry reduction is a constant-factor improvement, not an exponential simulation theorem. The number |V_m| of legal conformations still grows exponentially with chain length in this lattice

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representation. Efficient access to one ring address also does not mean efficient construction of all amplitudes in a generic quantum state. For arbitrary circuits, interference, non-Clifford resources, and entanglement structure can prevent tractable classical simulation. This paper makes no unconditional claim to match general-purpose quantum computation.

"Low power" must therefore become a measurement contract: energy per solved instance (joules), maximum resident memory, wall-clock time, numerical error, successful reachability coverage, and hardware and software versions. Compare the full sparse solver, sector solver, a conventional HP Monte Carlo baseline, and, where practical, a quantum-hardware or simulator baseline under the same input/output specification. No energy or power draw was measured in this study.

A practical classical quantum compiler would need to demonstrate not just a saving in nominal state count but an end-to-end advantage after constructing geometry, checking legality, building sparse operators, propagating the desired times, and reconstructing measured observables. The existing paired-sector test validates numerical identity. It supplies no electrical energy measurements, quantum-hardware benchmark, or asymptotic claim. Those distinctions should remain visible in any abstract, presentation, or downstream AI application.

PART V / WHAT THE TWO INVESTIGATIONS ESTABLISH TOGETHER

16. The missing physical interface is a typed constraint

The binary route defines lattice conformation unambiguously; the local hexagon move defines a geometrically legal endpoint pair. Neither fact proves that a peptide can continuously execute the move through collision-free three-dimensional configurations, nor determines its rate in solvent. The HP energy counts lattice contacts, not entropic free energy, dihedral barriers, solvent interactions, or chaperone coupling. Real folding kinetics may require environmental degrees of freedom and thermodynamic irreversibility at the observed subsystem level.

The construction $H=V-\gamma A$ is a legitimate Hermitian generator on the declared conformation graph; complex amplitudes show interference in that mathematical model. Its amplitude interpretation as an actual molecular wavefunction is not yet justified. In particular, a mathematically nontrivial Z_2 sign on a quotient loop is not a measured Berry phase or a molecular magnetic flux. Likewise, the ring's twisted clock is an exact reversible permutation but is not, by itself, a derived physical Hamiltonian.

For the scientific objective "predict a folding mechanism," the missing interface is now precisely typed: a lawful microscopic or coarse-grained transition kernel that maps real molecular coordinates and environmental variables to the accepted graph edges, rates, and phases. It must preserve its inverse on the closed enlarged system while permitting an effectively dissipative reduced observation. Assigning γ uniformly to every legal edge is not a derivation of that kernel. Adding this operator by convention and then calling the simulation a landing would violate the compiler-first method.

Write E for an encoding of molecular configurations into a computational Hilbert space, and let H_phys be the molecular generator under declared environmental conditions. A successful native physical correspondence would need to establish the same predictions in both descriptions—not just an address

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bijection. One mathematically sharp condition, where the domains permit it, is an intertwining of generators:

E H_phys = H_comp E (or an explicit open-system/channel analogue).

That equation is not satisfied by definition. It names the missing physical link: the compiler must supply the same allowed transitions and phases as the molecule. Exact ring permutation identities and finite-unitary simulation do not furnish H_phys. When the system is open, the correct comparison uses the larger system-plus-environment unitary or a reduced quantum channel with specified conditions.

The two manuscripts therefore address different stages of one dependency chain. In the square toy, a convenient address successor was improperly promoted to a molecular move. In the honeycomb model, state legality and reversible edge generation are derived from an explicit geometry, eliminating padding leakage. The remaining question has moved outward to physical admissibility: continuous collision-free paths, molecular force fields, rates, coherent or dissipative interactions, and temperature/solvent dependencies.

Required link	What the uploads already supply	What remains OPEN
Configuration legality	Six exact cyclic 2D conformations; valid projector P	A 3D, chemically appropriate state domain
Transition grammar	Native successor, shifts, and reversible inversion	Molecule-derived allowed local or concerted moves
Transition generator	Assumed cyclic K and HP potential V	Rates/amplitudes/phases from physical interaction
Environmental return	Exact reversible dilation existence	Coupling to solvent, heat and chaperones
Constraint preservation	[H,P]=0 criterion; PKP example	A connected, physically allowed mixer
Quantum readout	Amplitudes, QFT, Strang convergence	Independent observables and folding kinetics
Multi-compiler scaling	Odd-lane locks; temporal reader; deviation transport	A lawful molecular-to-lane encoding and update interface
AI exact queries	Inward/outward address and fully specified readouts	Task-specific correctness and verified operator contracts

Interface inventory: exact mathematical provision versus molecular or computational link still required.

17. Relation to established computational and biological work

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Quantum lattice-protein work predates this construction. Perdomo-Ortiz and collaborators used quantum annealing for low-energy protein conformations [5]; Robert and collaborators constructed a resource-aware variational quantum model for protein folding [10]. Those studies principally address energy optimization or sampled low-energy conformations, whereas the present analysis focuses on route legality and exact parity-resolved execution of a chosen dynamical generator. None of these approaches by itself yields a validated continuous-time biochemical folding mechanism.

The Gottesman–Knill theorem establishes that a rich, but restricted, quantum circuit family is efficiently simulable classically [6, 11]. Quantum-inspired results by Tang and others establish further dequantizations under explicit low-rank/sample-access assumptions [7, 8]. Nexus does not supersede these results; a useful differentiating claim will require either a new provable problem class with asymptotically better classical cost, or a reproducible favorable accuracy/time/energy benchmark against strong classical baselines. Here the distinct compiler contribution is the integration of native reversible return paths, a geometry-generated lawful edge set, and an exact symmetry-sector projection.

Turn encodings and folding-energy optimization have substantial prior art. The distinct content established here is the joint audit of address-versus-geometry errors, an exact native inverse for the legal route, a C-equivariant Hamiltonian built on that route graph, and numerical verification of an independently compiled parity block on classical hardware. It should be described as a structured simulation mechanism for this finite model, not a proof that all protein folding or general quantum computation is efficiently classical.

18. Correction ledger and the scope of each claim

"Defect" is a reference-relative deviation, not a faulty word; closed reversible evolution transports it rather than automatically healing it.

Native adjacency is an address relationship; it is not a molecular move until an independent law supplies that equivalence.

A ring permutation can be implemented as a unitary, but coherent phase dynamics and thermodynamic folding are additional operations.

High penalties suppress invalid population; they do not enforce strict legality at finite Λ. Deleting padding also deletes virtual bridges.

A perfect route home does not imply compression: generic n-bit words retain n-bit route information, while exact locked families have fewer independent degrees of freedom.

The center returns an innermost shell difference; interpreting it as semantic yes/no requires a declared question.

One Inverter's even/odd holonomy and ring-sphere constructions are exact finite algebra/topology, not themselves evidence of molecular geometry.

Symmetric potential levels do not prescribe rates between them; the three reflection pairs of PHPPHP are energy-equivalent, not automatically dynamically linked.

Audited states: STANDS, RUNS, and OPEN

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Claim	Status	Precise scope
Partner, inverse E/G route, twisted clock identities	STANDS	Defined discrete register mathematics
Center trace parity, odd-lane lift and partner intertwining	STANDS	Specified ring and reader
Unitary permutation lift UF; invariant-subspace criterion	STANDS	Standard finite-dimensional linear algebra
Six PHPPHP conformations and pairwise geometry counts	RUNS	Uploaded cyclic report; all 15 pairs
Four padding-crossing edges; coefficient 49/75	RUNS	Exact 8×8 Hamiltonian; gamma=0.7
Projected valid graph two P_3 components; contact cap 1/2	RUNS	Given initial uniform six-state vector
Penalty leakage, commutators, Strang errors	RUNS	Specified toy potential/hopping; dimensionless t=1
Molecule-derived geometry-to-transition operator	OPEN	No molecular move law in the supplied ring Hamiltonian
Dissipative folding and bath/channel kinetics	OPEN	No environmental coupling in supplied simulation
Generality, physical quantum mechanism, AI advantage	OPEN	No independent out-of-sample physical validation

Research-status ledger preserved from the first compiler-side audit.

Additional STANDS from the honeycomb analysis are the free complement involution, the exact parity block decomposition for C-stable declared Hamiltonians, and the signed-sheet return rule. The honeycomb graph counts and parity propagation RUNS at the enumerated lengths and times. The all-length connectivity of particular local move sets, physical realization of their rate law, and net power improvement remain OPEN. The independent §146 follow-up should be cited for the distinction between binary codimension and geometric disconnection.

19. Exact next computational and experimental specification

The next required WHAT is not a different numerical value or a new ring size. It is a valid molecular transition relation G_phys on a specified conformation space. Every edge must be justified by an allowed transformation of residue coordinates or molecular degrees of freedom; it must carry an inverse or an

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explicit environment record; and the corresponding Hamiltonian or channel must satisfy the legality projection. Only then should energy phases and time evolution be run.

The sealed input should specify sequence, spatial/chemical environment, configuration variables, admissibility constraints, transition verb, operator generator, and a measurable readout. It should predict a result before comparison with external folding data. If the relation cannot be constructed, the status remains OPEN and the missing provider is G_phys—not a missing index in the eight-address cycle.

The next prediction must have a domain-side input rather than an inferred answer. Fix a molecular sequence and a conformation representation with explicitly checked admissibility, identify a geometrically supported transition with a return record, derive or measure the rate/phase/environmental coupling rather than assigning all edges the same γ, and seal a measurable readout before linking an independent physical dataset. In parallel, benchmark a full sparse solver and a compiled symmetry solver with matched error tolerance, timing, memory, and electrical energy per solved instance. A speed or power claim belongs only to those measured comparisons.

The compiler proof obligation

Given E, H_phys and H_comp: certify E H_phys = H_comp E, or a stated open-system channel analogue.

Until that intertwinement or experimentally predictive substitute is supplied, the mathematical-space computation is RUNS, the physical protein-folding equivalence is OPEN, and the general low-power hypothesis remains an empirical target. That separation is not a weakening of the method; it is the dependency boundary the compiler has actually identified.

20. Conclusion

The work supports an operational principle: execute in the representation where the transformation is local, retain the way home, and read in the representation where the requested distinction is visible. Honeycomb turn coordinates give a geometric route; legal rewrites give reversible state transitions; mirror symmetry supplies a one-bit quantum Fourier split; the resulting parity blocks can be propagated on ordinary classical hardware without numerical disagreement from full-space propagation at the tested scale.

The negative controls carry equal weight. Address adjacency was not a molecular move. Short-range changes fragmented the graph. The GF(2) deficiency did not count geometric components. The parity reduction halved a state vector only for a parity-selected preparation. None of these facts licenses a universal claim that classical computers efficiently replace quantum hardware. They do define a concrete and falsifiable research program for quantum-compatible mathematical compilation without quantum processors in restricted, physically motivated domains.

The native compiler mathematics supplies exact routes, inverses, local observation rules, and constrained state families. Quantum mechanics supplies an exact unitary representation of reversible transformations and a necessary accounting for hidden return information. The present folding toy supplies an explicit energy catalog and chosen coherent evolution; its kinetic coupling is not yet produced from the molecule. The first unsupplied provider is the molecular transition grammar, and its absence explains both padding leakage and disconnected valid-state dynamics. The logic therefore identifies the next required compiler

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interface without inventing a physical fold. The mathematics RUNS; the proposed physical equivalence remains OPEN.

The integrated result is therefore a complete finite example of constraint-first quantum-style computation in mathematical state space: geometry supplies states and transitions, reflection supplies an exact representation split, and the reduced quantum Hamiltonian runs on standard processors with an explicit route home. The first model explains what fails when address adjacency impersonates molecular motion; the second supplies an exact correction for the toy honeycomb domain. The next leap is not another address table. It is a physically justified transition generator and a benchmark that measures power, time, memory, and out-of-sample predictive behavior.

APPENDICES / DERIVATIONS, PROVENANCE AND REFERENCES

Appendix A. Core proofs in one place

A.1 Permutation-unitary lift. For a bijection F on a finite set V, the matrix U_F has entries (U_F)_{y,x} = 1 if y = F(x), otherwise 0. Every row and every column contains exactly one nonzero entry. Hence U_F†U_F = I. This theorem establishes a valid unitary representation of the declared permutation; it does not select any Hamiltonian for a molecule.

A.2 Valid-domain invariance. Let P be an orthogonal projector and Q = I − P. For Hermitian H, [H,P] = 0 if and only if QHP = 0, because PHQ = (QHP)†. The resulting block-diagonal H preserves both sectors under exp(−iHt). For an initial P-state ψ_0, the leading invalid amplitude under a non-invariant H is $-itQH\psi_0$, yielding $t^2\|QH\psi_0\|^2 + O(t^3)$ for the probability. In the eight-state toy, $\|QK\psi_0\|^2 = 49/75$.

A.3 Parity decomposition. A fixed-point-free complement C partitions legal states into two-element orbits {a,Ca}. If V(Ca)=V(a) and edge multiplicities satisfy A(Cx,Cy)=A(x,y), then [H,U_C]=0. In the ordered pair basis, U_C acts as a Pauli X on the face degree of freedom. Applying a two-state Hadamard / Fourier transform diagonalizes U_C. The even and odd blocks have adjacency A_0+A_1 and A_0-A_1. The inverse transform reconstructs ψ_a and ψ_Ca from ϕ_+ and ϕ_− without dropping the face distinction.

A.4 Holonomy of the signed quotient. Fix an arbitrary representative of each complement orbit. Mark each lifted transition σ_e = +1 if it lands on the same representative sheet and −1 if it crosses. Changing representatives flips incident edge signs, but the product along a closed route remains unchanged. A −1 product identifies a lifted route ending at the complement of its start. Whether such a loop exists is a fact about the selected legal graph, not about the abstract complement involution alone.

A.5 Exact molecular return and environment. Suppose unitary U maps $|i\rangle|0\rangle$ to $|g\rangle|e_i\rangle$ for distinct orthonormal $|i\rangle$. Inner-product preservation requires $\langle e_i|e_j\rangle = \delta_{ij}$. A subsystem may thus look convergent while the larger system retains the inverse. This algebra requires no claim that the simple SWAP implementation reproduces thermodynamic folding.

A.6 Cost boundaries. One parity block uses N/2 complex amplitudes when the initial state and the declared Hamiltonian preserve parity. A generic state occupies both sectors and retains N complex degrees of freedom. The present reduction is constant-factor; arbitrary-state exponential simulation complexity is not ruled out. The hamiltonian spectrum, legal graph, and energy observable depend on their declared source and must not be inferred from native address geometry.

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A.7 Mirror lamps, temporal return, and embedding contract. For n=2h, put $D_i=x_i \oplus x_{i+1}$ with i modulo n. A complete τ-orbit of the center reads the cyclic D_i in known order, except for one inverted boundary crossing. Their XOR must therefore equal one. The D_i determine x up to global complement because the only kernel of the cycle-edge difference map consists of the two constant words. The retained native face bit resolves the missing complement. For the odd-lane lift, reversing lane order preserves the alternating-complement arrangement and synchronizes the conditional palindrome branch of P. This proves the stated operator intertwining, not a corresponding identity for T.

A.8 The local-return obstruction. On a honeycomb embedding, two alternative fixed-endpoint paths differing at just one interior site would create a four-cycle, forbidden by the honeycomb girth of six. Two adjacent changed sites yield the unique six-cycle exchange. Any new three-site alternative would require a distinct eight-cycle, but the honeycomb plane has none. The next planar boundary consists of two adjacent six-sided faces sharing an edge: its outer perimeter has 6+6−2=10 edges. Two opposite five-edge arcs therefore give a four-site geometric return. This proves the minimum geometrically distinct bounded-support exchange, not that all its atomistic physical realizations are possible.

Appendix B. Input data, numerical checks, and source distinction

Primary provided sources: "One_Inverter.docx" (NEXUS One Inverter, October 2026); "NEXUS_ledger_and_audit_1-147.md" (§§135–147); "Pasted markdown (3).md" (user instruction and initial HPPH quantum study); "Pasted text (2)(1).txt" (cyclic PHPPHP script and results); "Pasted text(9).txt" (initial quantum study); "quantum_ring_fold_bundle.zip" (source, traces, matrix package, error table); "cyclic_report.json" and "cyclic_matrices.npz" (six-residue run); associated plots supplied with the files.

New independent checks were executed with "NEXUS_quantum_compiler_constraint_audit.py" using the supplied coordinate lists and eight-state report. The script prints its prediction text hash before data loading and lists each finite check. It does not qualify as an external time-stamp or an unseen-data experimental prediction, since the corpus was available in the present session. Reported values not independently rerun are identified as source-reported; the printed leakage/geometry/commutator/Strang values were independently reproduced.

Independent audit prediction SHA-256:
0f60cd3e821b0a3fe7f16e78a08540e8d54f45dbfedc0a18726f1e68469a8075

Geometry (15 unordered pairs): distances by number of changed residue coordinates {3:6, 4:9}; no single-site edges. Mirrored conformation pairs: (0,1), (2,5), (3,4). Address edges crossing padding: (2,3), (3,4), (6,7), (7,0). Projected valid edges: (0,1), (1,2), (4,5), (5,6). Leakage early-time coefficient: 49/75. $\|QKP\|_2$: 0.989949493661. Effective eliminated-padding bridges: (0,6) and (2,4), each with H_PQ H_QP coefficient 0.49. The environment SWAP route passes a six-state orthogonality-and-return test.

For the Strang simulation using V=[−1,−1,0,10,0,0,0,10], $K=-0.7(T_8+T_8^\dagger)$, the independently recomputed error series at r=(4,8,16,32,64,128) was (0.082770330, 0.018193978, 0.004416348, 0.001096147, 0.000273545, 0.000068356).

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The independent audit is supplied as NEXUS_quantum_classical_sector_audit.py with its printed output and JSON results. The prediction and return route were committed in the source before the geometry and sparse-matrix calculations. Its SHA-256 seal is:

aa07c86a03b3f538c0472520707a5a2f85d940d55e0066959eb9873a4a062557

Execution environment: Python 3, NumPy, SciPy, integer honeycomb coordinates, sparse CSR Hamiltonians and SciPy expm_multiply. The supplied NEXUS_quantum_classical_sector_audit.py is bundled with the revised manuscript. It enumerates all binary m-turn words at m = 8 and m = 10, filters invalid chains, constructs legal edge families, verifies complement edge closure, assigns the declared HP potential, independently constructs full and parity-reduced Hamiltonians and checks numerical return. Its source preparation for m = 8 is 00001000; for m = 10 it is 0000100010. Additional NEXUS_147_figure_results.json and dense-contact CSV regenerate Figures 6 and 7; these are post-review diagnostics and are not represented as a new pre-data SHA seal.

The §146 locality code and Claude's independent cross-check establish the graph vocabulary, mod-four legal paths, and the both-end row. Claude's §147 verification also independently reproduced all square-toy values and retested full-versus-parity evolution using different numerical solvers. Its eigendecomposition crossed the presealed 5×10^{-16} tolerance at 1.7×10^{-15}; this numerical check is a recorded FAIL at that threshold, not a failure of algebraic block equivalence. No experiment in this record measures joules or validates protein folding kinetics.

Provenance notation. Numerical results from the square-lattice audit are preserved as the first report's independently reproduced values; the graph counts in §§145–147 are sourced from the project audit and reproduced notebook; the new classical parity-sector comparison is the second report's sparse-matrix run. The present integrated manuscript reorganizes those reported checks but does not claim a fresh, unseen physical experiment or a new sealed prediction. The source materials are the two uploaded October 23 manuscripts and their cited supporting code and ledger.

Additional §148 audit. The source NEXUS_148_local_return_gate_audit.py committed a prediction, including the inverse arc contract and explicit UNLINKED test for gate unlocking, before running the geometry. Prediction SHA-256: 00137ca45664f8039f645bb37d1fd959067bd98fc93b3a31b3652b2bf45c76ff. The executed report is NEXUS_148_local_return_gate_output.txt; independent post-seal graph and route computations are NEXUS_148_extended_route_reach.py and NEXUS_148_return_path_witness.py, with printed outputs. The §147 paper audit and its solver-dependence correction remain unchanged. Section numbers in the historical project ledger are reported with their original provenance; this paper incorporates a new §148 calculation beyond that prior ledger.

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